25900451

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCHTC200 - FLORICULTURE AND OLERICULTURE
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **1** mark

1. What is a bouquet? [U] / [CO2]
   A bouquet is an arrangement of cut flowers artistically grouped together.

2. What is the proper method for storing Cucumber? [K] / [CO3]
   Cucumbers should be stored at 10–13°C and high humidity to prevent chilling injury. Proper refrigeration in normal climate is sufficient.

3. Which cut flower is most popular in the global market? [K] / [CO2]
   Rose.

4. What are the two main types of components that nursery media mainly consist of? [K] / [CO2]
   Organic and inorganic components. (**Give marks if any examples of organic and inorganic components are written**).

5. What is the typical crop duration for Tuberose? [K] / [CO2]
   6–8 months.

6. Name any two plant growth regulators. [K] / [CO3]
   Auxins and Gibberellins are examples of plant growth regulators. (**Or any other PGR**)

7. What are cool season vegetables? [U] / [CO3]
   Cool-season vegetables grow best in low temperatures, e.g., cabbage and carrot.

8. Which cultivation is promoted through rose shows? [K] / [CO1]
   Rose shows promote commercial rose cultivation and hybridization.

9. What is the scope of protected cultivation in floriculture? [U] / [CO2]
   Protected cultivation enables year-round flower production and quality improvement.

10. Name any two value-added products made from tomatoes. [K] / [CO3]
    Tomato ketchup and sauce are common value-added products. (**Any two relevant value-added products**)

11. In which country did floriculture as a commercial industry first develop?[K] / [CO1]
    Netherlands.

12. What is the most important method to prevent diseases in mushroom cultivation? [U] / [CO4]

Maintaining hygiene and sterilized condition. (Any method that comes under these heads)

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Discuss the environmental benefits of using paddy straw in mushroom farming. [U]/[CO4]

Paddy straw is biodegradable, improves aeration, substrate structure, retains moisture, and enhances mycelial growth in mushrooms, reducing pollution from burning.

14. Describe five practices to enhance soil health in kitchen gardens. [A] / [CO3]

Practices include adding compost, green manuring, crop rotation, mulching, and minimal chemical use to enhance soil fertility and microbial activity. **(Any relevant points)**

15. Discuss the relevance of seasonal vegetable farming under changing climate. [U] / [CO3]

Seasonal vegetable farming maintains yield stability, reduces pest incidence, conserves water, and ensures sustainability under climate variability.

16. Write a note on traditional containers in Indian floriculture. [U] / [CO2]

Description on earthen pots, bamboo baskets, and coconut shells, which promote eco-friendliness, cost-effectiveness, and local cultural aesthetics.

17. Discuss the scope of vegetable cultivation in urban areas. [U] / [CO3]

Provides fresh vegetables, utilizes limited spaces (terraces/balconies), enhances greenery, and supports food security. **(Any relevant points)**

18. Explain the role of the National Horticulture Board (NHB) in floriculture development. [U]/CO1]

NHB supports floriculture through subsidies, training, infrastructure development, and promotion of export-oriented production. **(Liberal valuation)**

19. Write a note on line arrangement. [U] / [CO2]

Line arrangement involves placing flowers in a linear design to emphasize direction and movement, common in modern floral design styles.

20. Briefly explain "Precision Planting" and "No-Till Planting" as modern planting methods and discus the benefits of each of them. [U] / [CO2]

Precision planting ensures uniform spacing and depth using mechanized tools, while no-till planting conserves soil moisture and structure by avoiding ploughing. **(Liberal valuation. Give marks if planting methods are described) (5X6=30)**

Part C

Essay

Answer any **1** question

Each question carries **10** marks

21. Write an essay on physiological and pathological factors affecting vase life of cut flowers. [U] / [CO2]

- Water stress and poor water uptake
- High respiration and transpiration rates
- Ethylene production and sensitivity
- Nutrient deficiency in vase solution
- Mechanical injury during harvest/transport
- Microbial contamination of vase water
- Improper temperature and humidity during storage
- Use of floral preservatives extends vase life
- Varietal differences and maturity stage affect longevity
- Fungal and bacterial infections cause early senescence

22. Discuss the role of national and state-level agencies in promoting vegetable and flower cultivation in India. [U] / [CO1]

- National Horticulture Board (NHB): Infrastructure support, training, and subsidies.
- National Horticulture Mission (NHM): Promotes integrated development and farmer training.
- APEDA: Encourages export of flowers and vegetables.
- ICAR & State Agricultural Universities: Conduct research and extension activities.
- State Departments of Agriculture/Horticulture: Implement schemes and distribute quality planting materials.
- KVKs: Provide on-field demonstrations and capacity building.
- NABARD: Offers financial support for agri-horticultural enterprises.
- MIDH (Mission for Integrated Development of Horticulture): Focuses on protected cultivation and post-harvest management.
- State Horticulture Missions: Promote region-specific crops and technologies.
- NGOs & Co-operatives: Help with marketing, awareness, and farmer organization.

(1X10=10)